ModelAudit — Summary (page 1)


```
{
  "dataset": "taiwan",
  "model": "xgb",
  "scope": "research demonstration only; thresholds are local review heuristics",
  "evidence_sha256": "8f4306268f09b6363563ecce7fc14e27685396ba0b259107b3a9051d75b0ecf3",
  "findings": [
    {
      "test_id": "STAB-CSI-01",
      "severity": "high",
      "confirmation": "reproduced"
    },
    {
      "test_id": "FAIR-AGE-01",
      "severity": "medium",
      "confirmation": "not_reproduced"
    },
    {
      "test_id": "ROB-MONO-01",
      "severity": "high",
      "confirmation": "reproduced"
    },
    {
      "test_id": "DQ-CAT-01",
      "severity": "high",
      "confirmation": "reproduced"
    }
  ]
}
```

ModelAudit — PERF-AUC-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "auc": 0.7908455754512915,
    "auc_ci95": [
      0.7768547465610005,
      0.8062859246403715
    ],
    "ks": 0.4469432581086302,
    "gini": 0.581691150902583,
    "count": 6000
  },
  "threshold": {
    "metric": "auc",
    "warn_below": 0.75,
    "fail_below": 0.7
  }
}
```

ModelAudit — CAL-ECE-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "brier": 0.12744802315090895,
    "ece": 0.023504875183105463,
    "calibration_in_the_large": -0.002154301047325141,
    "hosmer_lemeshow_stat": 36.96525873108081,
    "hosmer_lemeshow_df": 8,
    "hosmer_lemeshow_p": 1.1679467296849143e-05,
    "reliability_bins": [
      {
        "bin": 0,
        "count": 600,
        "observed_rate": 0.03666666666666667,
        "predicted_rate": 0.07143252690633138
      },
      {
        "bin": 1,
        "count": 600,
        "observed_rate": 0.065,
        "predicted_rate": 0.09205266316731771
      },
      {
        "bin": 2,
        "count": 600,
        "observed_rate": 0.085,
        "predicted_rate": 0.10535832087198893
      },
      {
        "bin": 3,
        "count": 600,
        "observed_rate": 0.09666666666666666,
        "predicted_rate": 0.11927978515625
      },
      {
        "bin": 4,
        "count": 600,
        "observed_rate": 0.13166666666666665,
        "predicted_rate": 0.13519869486490885
      },
      {
        "bin": 5,
        "count": 601,
        "observed_rate": 0.1464226289517471,
        "predicted_rate": 0.15573381623888571
      },
      {
        "bin": 6,
        "count": 599,
        "observed_rate": 0.17696160267111852,
        "predicted_rate": 0.18762668106512156
      },
      {
        "bin": 7,
        "count": 600,
        "observed_rate": 0.255,
```

ModelAudit — CAL-ECE-01 (page 2)


```
        "predicted_rate": 0.24171216328938802
      },
      {
        "bin": 8,
        "count": 600,
        "observed_rate": 0.39166666666666666,
        "predicted_rate": 0.3649652608235677
      },
      {
        "bin": 9,
        "count": 600,
        "observed_rate": 0.725,
        "predicted_rate": 0.6582364908854167
      }
    ]
  },
  "threshold": {
    "metric": "ece",
    "warn_above": 0.03,
    "fail_above": 0.05
  }
}
```

ModelAudit — STAB-PSI-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "psi": 0.01343728427133399,
    "baseline_test_psi": 0.01721937212872707,
    "scenario": "limit -20% and age +5"
  },
  "threshold": {
    "metric": "psi",
    "warn_above": 0.1,
    "fail_above": 0.25
  }
}
```

ModelAudit — STAB-CSI-01 (page 1)


```
{
  "status": "fail",
  "values": {
    "max_csi": 0.29407266523914694,
    "feature": "LIMIT_BAL",
    "baseline_test_csi": 0.012817605796263731
  },
  "threshold": {
    "metric": "max_csi",
    "warn_above": 0.1,
    "fail_above": 0.25
  }
}
```

ModelAudit — FAIR-SEX-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "threshold": 0.25,
    "groups": {
      "female": {
        "approval_rate": 0.7847725774555043,
        "good_outcome_tpr": 0.8671556642216789
      },
      "male": {
        "approval_rate": 0.7454484153742414,
        "good_outcome_tpr": 0.8451754385964912
      }
    },
    "demographic_parity_difference": 0.03932416208126288,
    "equalized_odds_difference": 0.022212727455514225,
    "approval_rate_ratio": 0.9498910089229099,
    "overall_approval_rate": 0.7653333333333333,
    "approved_observed_default_rate": 0.11694250871080139
  },
  "threshold": {
    "metric": "equalized_odds_difference",
    "warn_above": 0.05,
    "fail_above": 0.1
  }
}
```

ModelAudit — FAIR-AGE-01 (page 1)


```
{
  "status": "warn",
  "values": {
    "threshold": 0.25,
    "groups": {
      "35plus": {
        "approval_rate": 0.7646664431780087,
        "good_outcome_tpr": 0.8546160483175151
      },
      "under35": {
        "approval_rate": 0.7659927079880676,
        "good_outcome_tpr": 0.8584437086092715
      }
    },
    "demographic_parity_difference": 0.0013262648100589303,
    "equalized_odds_difference": 0.05678505748564422,
    "approval_rate_ratio": 0.9982685673163358,
    "overall_approval_rate": 0.7653333333333333,
    "approved_observed_default_rate": 0.11694250871080139
  },
  "threshold": {
    "metric": "equalized_odds_difference",
    "warn_above": 0.05,
    "fail_above": 0.1
  }
}
```

ModelAudit — ROB-MONO-01 (page 1)


```
{
  "status": "fail",
  "values": {
    "test_feature": "PAY_0 missed-payment months",
    "violations": 64,
    "sample_rows": 400,
    "max_absolute_change": 0.35419997572898865,
    "missing_predictions_finite": true
  },
  "threshold": {
    "metric": "violations",
    "warn_above": 0,
    "fail_above": 1
  }
}
```

ModelAudit — DQ-CAT-01 (page 1)


```
{
  "status": "fail",
  "values": {
    "rows": 6000,
    "null_cells": 0,
    "duplicate_feature_rows": 2,
    "invalid_categories": 103,
    "label_out_of_range": 0,
    "id_and_target_excluded": true
  },
  "threshold": {
    "metric": "invalid_categories",
    "warn_above": 0,
    "fail_above": 100
  }
}
```

ModelAudit — FAIR-TRADEOFF (page 1)


```
{
  "status": "information",
  "values": {
    "thresholds": [
      {
        "threshold": 0.15,
        "groups": {
          "female": {
            "approval_rate": 0.5392221489782465,
            "good_outcome_tpr": 0.6157294213528932
          },
          "male": {
            "approval_rate": 0.5161834120026972,
            "good_outcome_tpr": 0.6035087719298246
          }
        },
        "demographic_parity_difference": 0.023038736975549257,
        "equalized_odds_difference": 0.012220649423068664,
        "approval_rate_ratio": 0.9572741271492564,
        "overall_approval_rate": 0.5278333333333334,
        "approved_observed_default_rate": 0.0884117461319861
      },
      {
        "threshold": 0.2,
        "groups": {
          "female": {
            "approval_rate": 0.6941331575477917,
            "good_outcome_tpr": 0.7836185819070904
          },
          "male": {
            "approval_rate": 0.667565745111261,
            "good_outcome_tpr": 0.7653508771929824
          }
        },
        "demographic_parity_difference": 0.026567412436530713,
        "equalized_odds_difference": 0.027048356288328135,
        "approval_rate_ratio": 0.961725769547752,
        "overall_approval_rate": 0.681,
        "approved_observed_default_rate": 0.10230053842388644
      },
      {
        "threshold": 0.25,
        "groups": {
          "female": {
            "approval_rate": 0.7847725774555043,
            "good_outcome_tpr": 0.8671556642216789
          },
          "male": {
            "approval_rate": 0.7454484153742414,
            "good_outcome_tpr": 0.8451754385964912
          }
        },
        "demographic_parity_difference": 0.03932416208126288,
        "equalized_odds_difference": 0.022212727455514225,
        "approval_rate_ratio": 0.9498910089229099,
        "overall_approval_rate": 0.7653333333333333,
```

```
        "approved_observed_default_rate": 0.11694250871080139
      },
      {
        "threshold": 0.3,
        "groups": {
          "female": {
            "approval_rate": 0.8174027686222808,
            "good_outcome_tpr": 0.8960880195599022
          },
          "male": {
            "approval_rate": 0.7933243425488874,
            "good_outcome_tpr": 0.8881578947368421
          }
        },
        "demographic_parity_difference": 0.02407842607339339,
        "equalized_odds_difference": 0.007930124823060103,
        "approval_rate_ratio": 0.9705427642311792,
        "overall_approval_rate": 0.8055,
        "approved_observed_default_rate": 0.12600869025450032
      },
      {
        "threshold": 0.35,
        "groups": {
          "female": {
            "approval_rate": 0.8651944627554383,
            "good_outcome_tpr": 0.9376528117359413
          },
          "male": {
            "approval_rate": 0.837491571139582,
            "good_outcome_tpr": 0.9241228070175439
          }
        },
        "demographic_parity_difference": 0.027702891615856373,
        "equalized_odds_difference": 0.013530004718397382,
        "approval_rate_ratio": 0.9679807340333302,
        "overall_approval_rate": 0.8515,
        "approved_observed_default_rate": 0.13720884713251125
      }
    ]
  },
  "threshold": null
}
```

ModelAudit — EXPL-SHAP-01 (page 1)


```
{
  "status": "information",
  "values": {
    "dataset": "taiwan",
    "rows_explained": 100,
    "xgb_shap_global_top10": [
      {
        "feature": "num__PAY_0",
        "mean_abs_shap": 0.5742000341415405
      },
      {
        "feature": "num__PAY_AMT2",
        "mean_abs_shap": 0.0914265364408493
      },
      {
        "feature": "num__PAY_5",
        "mean_abs_shap": 0.0797201469540596
      },
      {
        "feature": "num__PAY_AMT1",
        "mean_abs_shap": 0.07878690958023071
      },
      {
        "feature": "num__LIMIT_BAL",
        "mean_abs_shap": 0.07741788774728775
      },
      {
        "feature": "num__PAY_AMT3",
        "mean_abs_shap": 0.06772103905677795
      },
      {
        "feature": "num__PAY_2",
        "mean_abs_shap": 0.06638272851705551
      },
      {
        "feature": "num__BILL_AMT1",
        "mean_abs_shap": 0.06513115018606186
      },
      {
        "feature": "num__PAY_4",
        "mean_abs_shap": 0.05817516893148422
      },
      {
        "feature": "num__PAY_3",
        "mean_abs_shap": 0.05667481571435928
      }
    ],
    "xgb_first_case_adverse_top3": [
      {
        "feature": "num__PAY_0",
        "shap_log_odds": 1.6539591550827026
      },
      {
        "feature": "num__PAY_2",
        "shap_log_odds": 0.18914002180099487
      },
```

ModelAudit — EXPL-SHAP-01 (page 2)


```
      {
        "feature": "num__LIMIT_BAL",
        "shap_log_odds": 0.05639992654323578
      }
    ],
    "scorecard_first_case_reason_codes": [
      "PAY_0",
      "LIMIT_BAL",
      "PAY_AMT1"
    ],
    "scorecard_top3_overlap_under_1pct_amount_change": 0.9833333333333333,
    "warning": "SHAP is model explanation in log-odds space, not a causal reason or lawful adverse-action
  notice."
  },
  "threshold": null
}
```

ModelAudit — SURV-01 (page 1)


```
{
  "status": "information",
  "values": {
    "dataset": "UCI Taiwan; retrospective April\u2013September payment statuses",
    "people": 10000,
    "person_month_rows": 50973,
    "event_definition": "first recorded payment delay of at least two months",
    "events": 2801,
    "censored": 7199,
    "cox_concordance_train": 0.6520831166192114,
    "cox_hazard_ratios": {
      "LIMIT_BAL": 0.7486072184023408,
      "SEX": 0.9152345011329724,
      "AGE": 0.9504288600537097,
      "EDUCATION": 1.0427342693473656,
      "MARRIAGE": 0.9900729352017101
    },
    "ph_test_p": {
      "AGE": 0.8877453482639952,
      "EDUCATION": 0.04681646040104071,
      "LIMIT_BAL": 0.05282408133367339,
      "MARRIAGE": 0.9230135173119021,
      "SEX": 0.11013594798910528
    },
    "kaplan_meier": {
      "limit_below_100k": {
        "people": 3903,
        "survival_at_3": 0.7494235203689473,
        "survival_at_6": 0.5936459133999489
      },
      "limit_at_least_100k": {
        "people": 6097,
        "survival_at_3": 0.8807610300147616,
        "survival_at_6": 0.8007216663933082
      }
    },
    "discrete_hazard_train_auc": 0.7091065046785967,
    "warning": "Exploratory in-sample survival fit; event is delinquency proxy, not observed default. PH
  checks are diagnostics, not proof. Static default label is a different endpoint and later features can
  leak into early-event prediction."
  },
  "threshold": null
}
```